

ARITHMETIC FOUNDATIONS

NO CALCULATOR | 20 QUESTIONS | 44 RAW POINTS

V1

NAME	CLASS	DATE
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Show clear working. Give fractions in simplest form.

1

1 pt

Calculate $5/12 + 1/8$. Your answer is a/b in simplest form. Enter the numerator a .

ANSWER _____

2

1 pt

Calculate $8/15 \times 9/16$. Your answer is a/b in simplest form. Enter the denominator b .

ANSWER _____

3

1 pt

Calculate $-8.6 + 2.3$.

ANSWER _____

4

1 pt

Calculate 1.5×2 .

ANSWER _____

5

2 pts

To calculate $2/5 + 3/4$, what should you do first?

- A. Add the numerators and denominators separately
- B. Rewrite both fractions using a common denominator
- C. Multiply the fractions
- D. Flip the second fraction

6

2 pts

Evaluate $10 - 2(3 - 5)^2$.

ANSWER _____

7

2 pts

Evaluate $-2 + 3/4 \times 8$.

ANSWER _____

8

1 pt

Find 30% of 70.

ANSWER _____

9

2 pts

A price is increased by 15%. Which expression gives the new price if the original price is P ?

- A. $0.15P$
- B. $0.85P$
- C. $1.15P$
- D. $1.5P$

10

1 pt

A bag costs 50 USD and is discounted by 30%. Enter the new price in USD.

ANSWER _____

11

2 pts

Red and blue counters are in the ratio 4:7. There are 55 counters altogether. Enter the number of blue counters.

ANSWER _____

12

2 pts

A recipe uses 5 cups of flour for 12 muffins. How many cups of flour are needed for 36 muffins?

ANSWER _____

13

1 pt

Estimate 302×19.8 by using 300×20 . Enter the estimated product.

ANSWER _____

14

2 pts

To calculate $3/4 \div 2/5$, why can we multiply by $5/2$?

- A. Dividing by a fraction is equivalent to multiplying by its reciprocal
- B. Fractions must always have the same denominator
- C. Multiplication and division always give the same answer
- D. The larger fraction must come first

15

2 pts

Write $13/20$ as a decimal.

ANSWER _____

16

4 pts

An item has a list price of 160 USD. A sale takes 15% off the list price. A loyalty coupon then takes another 25% off the already discounted price. Finally, 13 USD shipping is added. Enter the final price in USD.

ANSWER _____

17

4 pts

Before anyone moves, the ratio of Group A to Group B is 5:7, and there are 120 people altogether. First, 20% of the people who started in Group B leave. Then 25% of the people still in Group B transfer to Group A. Enter the final number of people in Group B.

ANSWER _____

18

4 pts

A tank is $7/8$ full. Then $2/7$ of the water currently in the tank is used. The fraction of the whole tank remaining is a/b in simplest form. Enter the numerator a .

ANSWER _____

19

4 pts

Evaluate $5/6 \div 5/12 + 2(1 - 1/4)$. Enter your answer as a decimal.

ANSWER _____

20

5 pts

Fiction and nonfiction books are in the ratio 5:3. There are 64 books altogether. Then $1/5$ of the fiction books and 25% of the nonfiction books are sold. Enter the total number of books remaining.

ANSWER _____

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V2

NAME	CLASS	DATE
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Show clear working. Give fractions in simplest form.

1

1 pt

Calculate $7/15 + 1/6$. Your answer is a/b in simplest form. Enter the numerator a .

ANSWER _____

2

1 pt

Calculate $6/7 \times 14/15$. Your answer is a/b in simplest form. Enter the denominator b .

ANSWER _____

3

1 pt

Calculate $-7.4 + 3.1$.

ANSWER _____

4

1 pt

Calculate 1.2×3 .

ANSWER _____

5

2 pts

To calculate $3/8 + 5/6$, what should you do first?

- A. Flip the second fraction
- B. Multiply the fractions
- C. Add the numerators and denominators separately
- D. Rewrite both fractions using a common denominator

6

2 pts

Evaluate $12 - 3(4 - 6)^2$.

ANSWER _____

7

2 pts

Evaluate $-3 + 5/6 \times 12$.

ANSWER _____

8

1 pt

Find 25% of 84.

ANSWER _____

9

2 pts

A price is increased by 20%. Which expression gives the new price if the original price is P ?

- A. $0.20P$
- B. $0.80P$
- C. $1.02P$
- D. $1.20P$

10

1 pt

A bag costs 80 USD and is discounted by 25%. Enter the new price in USD.

ANSWER _____

11

2 pts

Red and blue counters are in the ratio 3:5. There are 64 counters altogether. Enter the number of blue counters.

ANSWER _____

12

2 pts

A recipe uses 4 cups of flour for 10 muffins. How many cups of flour are needed for 30 muffins?

ANSWER _____

13

1 pt

Estimate 398×31.2 by using 400×30 . Enter the estimated product.

ANSWER _____

14

2 pts

To calculate $2/3 \div 4/7$, why can we multiply by $7/4$?

- A. Fractions must always have the same denominator
- B. The larger fraction must come first
- C. Dividing by a fraction is equivalent to multiplying by its reciprocal
- D. Multiplication and division always give the same answer

15

2 pts

Write $7/20$ as a decimal.

ANSWER _____

16

4 pts

An item has a list price of 240 USD. A clearance sale takes 35% off the list price. A coupon then takes another 25% off the clearance price. Finally, 18 USD shipping is added. Enter the final price in USD.

ANSWER _____

17

4 pts

Before anyone moves, the ratio of Group A to Group B is 7:9, and there are 160 people altogether. First, 20% of the people who started in Group B leave. Then 25% of the people still in Group B transfer to Group A. Enter the final number of people in Group B.

ANSWER _____

18

4 pts

A tank is $5/6$ full. Then $2/5$ of the water currently in the tank is used. The fraction of the whole tank remaining is a/b in simplest form. Enter the numerator a .

ANSWER _____

19

4 pts

Evaluate $7/8 \div 7/16 + 5(1 - 1/2)$. Enter your answer as a decimal.

ANSWER _____

20

5 pts

Fiction and nonfiction books are in the ratio 3:2. There are 70 books altogether. Then $1/7$ of the fiction books and 25% of the nonfiction books are sold. Enter the total number of books remaining.

ANSWER _____

ARITHMETIC FOUNDATIONS

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V3

NAME	CLASS	DATE
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Show clear working. Give fractions in simplest form.

1

1 pt

Calculate $7/12 + 1/10$. Your answer is a/b in simplest form. Enter the numerator a .

ANSWER _____

2

1 pt

Calculate $10/21 \times 14/25$. Your answer is a/b in simplest form. Enter the denominator b .

ANSWER _____

3

1 pt

Calculate $-9.2 + 4.7$.

ANSWER _____

4

1 pt

Calculate 2.5×4 .

ANSWER _____

5

2 pts

To calculate $5/12 + 2/9$, what should you do first?

- A. Rewrite both fractions using a common denominator
- B. Flip the second fraction
- C. Add the numerators and denominators separately
- D. Multiply the fractions

6

2 pts

Evaluate $15 - 2(6 - 3)^2$.

ANSWER _____

7

2 pts

Evaluate $-4 + 2/5 \times 15$.

ANSWER _____

8

1 pt

Find 40% of 65.

ANSWER _____

9

2 pts

A price is increased by 8%. Which expression gives the new price if the original price is P ?

- A. $0.08P$
- B. $1.08P$
- C. $0.92P$
- D. $1.8P$

10

1 pt

A bag costs 60 USD and is discounted by 20%. Enter the new price in USD.

ANSWER _____

11

2 pts

Red and blue counters are in the ratio 5:4. There are 72 counters altogether. Enter the number of blue counters.

ANSWER _____

12

2 pts

A recipe uses 3 cups of flour for 8 muffins. How many cups of flour are needed for 40 muffins?

ANSWER _____

13

1 pt

Estimate 603×9.7 by using 600×10 . Enter the estimated product.

ANSWER _____

14

2 pts

To calculate $5/8 \div 3/10$, why can we multiply by $10/3$?

- A. Multiplication and division always give the same answer
- B. Dividing by a fraction is equivalent to multiplying by its reciprocal
- C. The larger fraction must come first
- D. Fractions must always have the same denominator

15

2 pts

Write $9/25$ as a decimal.

ANSWER _____

16

4 pts

An item has a list price of 180 USD. A sale takes 20% off the list price. A coupon then takes another 12.5% off the already discounted price. Finally, 19 USD shipping is added. Enter the final price in USD.

ANSWER _____

17

4 pts

Before anyone moves, the ratio of Group A to Group B is 4:7, and there are 176 people altogether. First, 25% of the people who started in Group B leave. Then 25% of the people still in Group B transfer to Group A. Enter the final number of people in Group B.

ANSWER _____

18

4 pts

A tank is $9/10$ full. Then $1/3$ of the water currently in the tank is used. The fraction of the whole tank remaining is a/b in simplest form. Enter the numerator a .

ANSWER _____

19

4 pts

Evaluate $2/3 \div 4/9 + 4(1 - 1/2)$. Enter your answer as a decimal.

ANSWER _____

20

5 pts

Fiction and nonfiction books are in the ratio 7:5. There are 72 books altogether. Then $1/7$ of the fiction books and 20% of the nonfiction books are sold. Enter the total number of books remaining.

ANSWER _____